

REACTR Validation Summary Table
Nathan Lee

	Documented phylogenetic tree gene family size (all species in paper)	Gene family size (only target and base)	% of documented orthologs also found by REACTR	False/Truncated isoforms detected	Robinson-Foulds distance of documented phylogenetic trees vs cleaned REACTR trees
DELLA: Vaccinium darrowii	39	8	100%	0	0
PHY: Oryza Sativa	20	8	100%	1	0
HOOKLESS: Solanum lycopersicum	8	4	100%	3	0

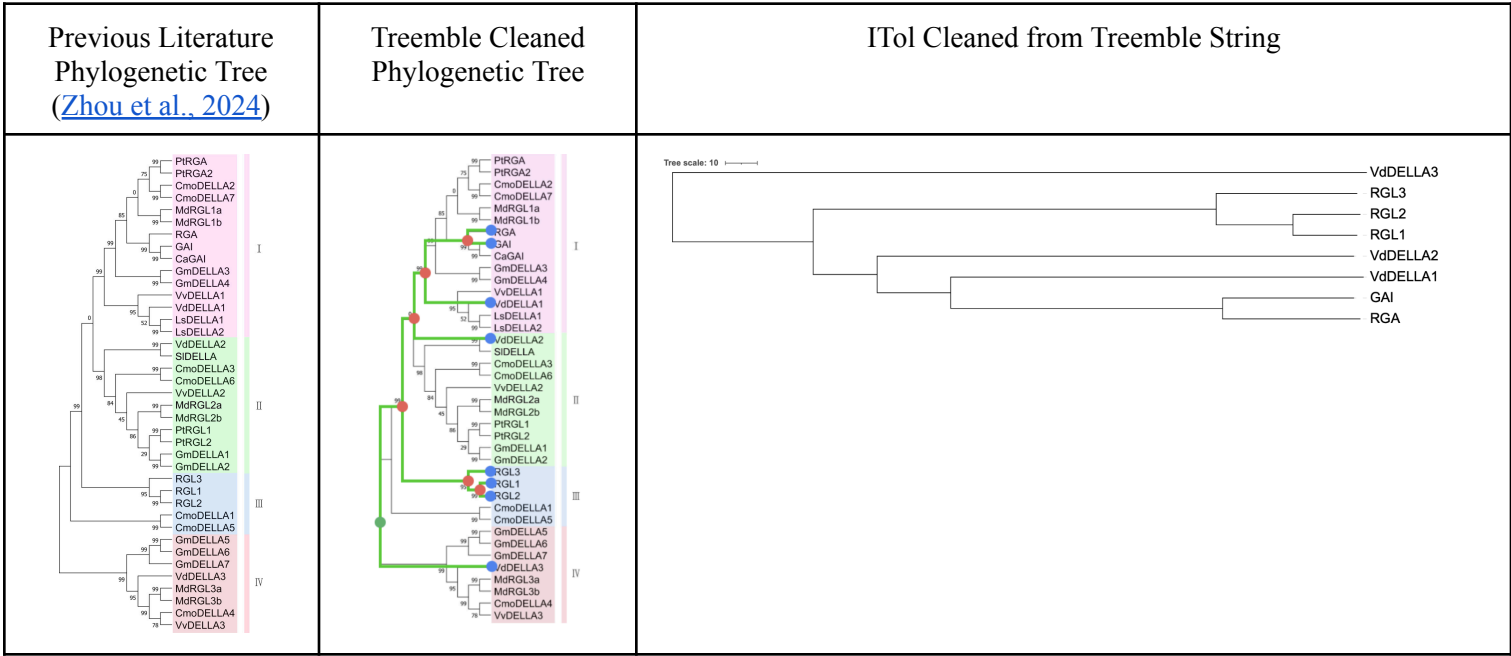
Methodology

REACTR was run on the base and target species, with the documented base species's gene family FASTAs as the query. The Newick string and image of the REACTR generated tree were downloaded. The locus tags of the protein sequences were then matched and renamed to fit the standardized protein symbols (i.e. RGL3) using UniProt ([The Uniprot Consortium, 2025](#)) and NCBI BLAST ([Altschul et al., 1990](#)) searches. The Newick tree was then cleaned to remove false/truncated isoforms.¹ Images of phylogenetic trees were downloaded from previous literature² The downloaded images were uploaded to Treemble ([Allard & Kumar, 2026](#)) to generate a Newick string. The Newick string was exclusively composed of the base and target species' gene families. The Newick string was uploaded to iTol ([Letunic & Bork, 2024](#)) to confirm that the Newick string was accurate to the phylogenetic tree from previous literature. The two Newick strings, from REACTR and previous literature (transformed through Treemble), were submitted to T-REX ([Boc et al., 2012](#)). The Robinson-Foulds calculated distance and its corresponding matrix was documented.

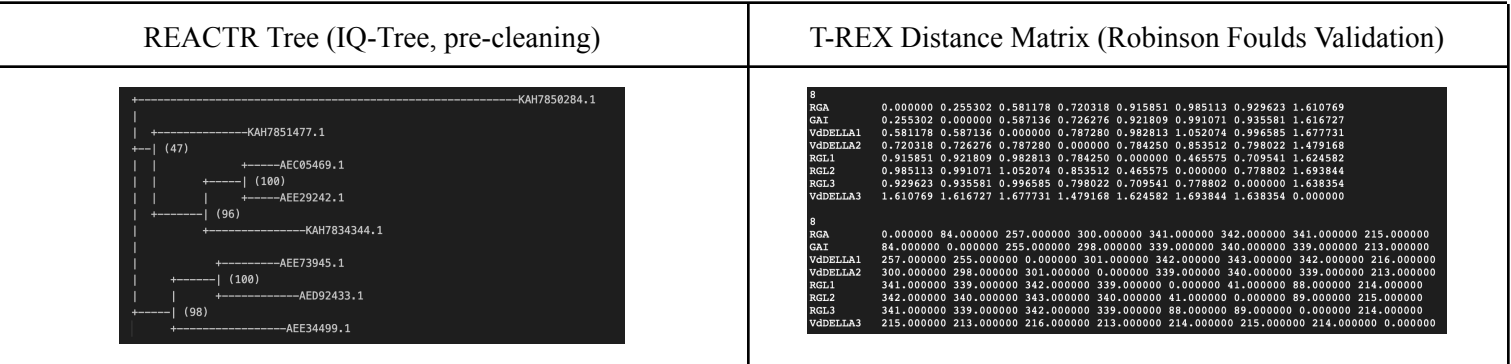
¹ False/truncated isoforms were removed for manual cleaning during validation. REACTR can pick up false isoforms during its homology-domain search, causing such isoforms to appear in downstream analysis like phylogenetic tree generation. Such isoforms are the effect of errors during genome assembly and reflect REACTR's sensitivity. However, isoforms can be removed manually through examination of phylogenetic trees. REACTR does not automatically remove isoforms due to a prioritization of sensitivity and emphasis on discovering complex, evolutionary events. Such designs are not uncommon amongst established tools, like Orthofinder ([Emms et al., 2026](#)).

² Retrieved from [Zhou et al., 2024](#) (DELLA: *Vaccinium darrowii*), [Schrager-Lavelle et al., 2016](#) (PHY: *Oryza Sativa*), [Chaabouni et al., 2016](#) (HOOKLESS: *Solanum lycopersicum*)

DELLA: *Vaccinium darrowii*



Treemle Generated Newick	REACTR Original Newick	REACTR Cleaned Newick (Tag names, identity matching)	Robinson-Foulds Distance (T-REX)
<pre>(((((RGA:43,GAI:41):85,VdDELLA1:129):23,VdDELLA2:149):20,(RGL3:44,(RGL1:20,RGL2:21):24):126):44,VdDELLA3:217):0;</pre>	<pre>(KAH7850284.1:1.1597499983,(KAH7851477.1:0.2943587276,((AEC05469.1:0.1246721418,AEE29242.1:0.1306299231)100:0.1324360415,KAH7834344.1:0.3240698106)96:0.1688512649)47:0.0250594701,((AEE73945.1:0.1981568522,AED92433.1:0.2674184850)100:0.1497274165,AEE34499.1:0.3616562861)98:0.1169479110);</pre>	<pre>(VdDELLA3:1.1597499983,(VdDELLA2:0.2943587276,((RGA:0.1246721418,GAI:0.1306299231)100:0.1324360415,VdDELLA1:0.3240698106)96:0.1688512649)47:0.0250594701,((RGL1:0.1981568522,RGL2:0.2674184850)100:0.1497274165,RGL3:0.3616562861)98:0.1169479110);</pre>	0



PHY: Oryza Sativa

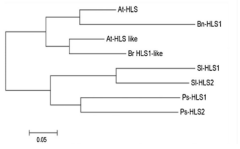
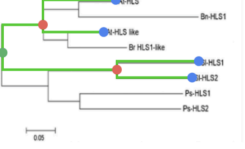
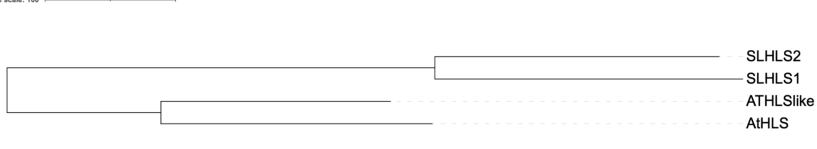
Previous Literature Phylogenetic Tree (Schrager-Lavelle et al. 2016)	Treemle Cleaned Phylogenetic Tree	ITol Cleaned from Treemle String

Treemle Generated Newick	REACTR Original Newick	REACTR Cleaned Newick	Robinson-Foulds Distance (T-REX)
<pre>((AtphyA:104,OsphyA:123):104,(OsphyC:89,AtphyC:168):57):46,(AtphyE:206,(OsphyB:80,(AtphyB:32,AtphyD:54):64):37):57):0;</pre>	<pre>(AEE84000.1:0.6545430041,(NP_001389072.1:0.1922894228,(AEE83721.1:0.1677459286,(ANM63140.1:0.0413747359,AEC06808.1:0.0000026845)100:0.0745601759)100:0.1360673694)100:0.1401141572,((NP_001405525.1:0.4283623775,AEE28462.1:0.2268402708)100:0.3323859187,(NP_001405645.1:0.2945056913,AED94021.1:0.4880709750)99:0.1326169304)100:0.3206583108);</pre>	<pre>(AtphyE:0.6545430041,(OsphyB:0.19,(AtphyB:0.17,AtphyD:0.0745601759)100:0.1360673694)100:0.1401141572,((AtphyA:0.4283623775,OsphyA:0.2268402708)100:0.3323859187,(OsphyC:0.2945056913,AtphyC:0.4880709750)99:0.1326169304)100:0.3206583108);</pre>	0

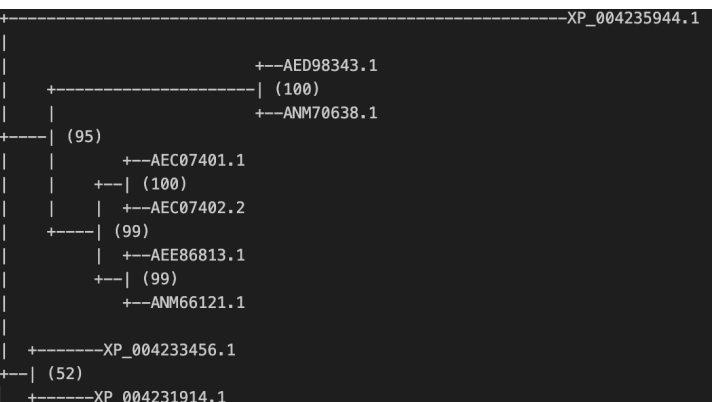
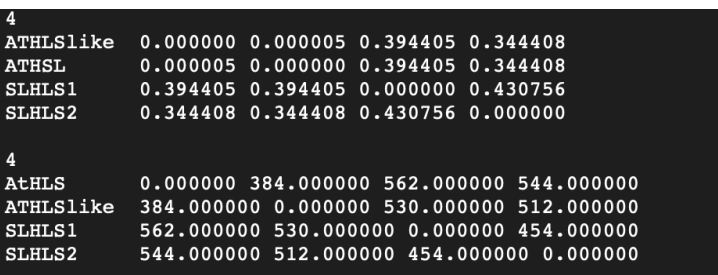
REACTR Tree (IQ-Tree, pre-cleaning)	T-REX Distance Matrix (Robinson Foulds Validation)

Removed isoform (ANM63140.1)

HOOKLESS: Solanum lycopersicum

Previous Literature Phylogenetic Tree (Chaabouni et al., 2016)	Treemle Cleaned Phylogenetic Tree	ITol Cleaned from Treemle String
		

Treemle Generated Newick	REACTR Original Newick	REACTR Cleaned Newick	Robinson-Foulds Distance (T-REX)
((AtHLS:208,ATHLSlike:176):118,(SLHLS1:236,SLHLS2:218):328):0;	(XP_004235944.1:1.6291598128,((AED98343.1:0.0000025602,ANM70638.1:0.0000021128)100:0.6049279491,((AEC07401.1:0.0000025602,AEC07402.2:0.0000021128)100:0.0970948845,(AEE86813.1:0.0000026968,ANM66121.1:0.0000021128)99:0.0669425103)99:0.1433370798)95:0.1540259248,(XP_004233456.1:0.2403762388,XP_004231914.1:0.1903798615)52:0.0797280129);	((ATHLSlike:0.0000025602,ATHSL:0.0000026968):0.1540259248,(SLHLS1:0.2403762388,SLHLS2:0.1903798615):0.0797280129);	0

REACTR Tree (IQ-Tree, pre-cleaning)	T-REX Distance
	

Removed three isoforms (ANM70638.1, AEC07402.2, ANM66121.1), removed extra paralog (AEC07401.1) to match query of reference literature